

Single-sided Multi-layer Electric Circuit by Hot Stamping with 3D Printer

Yuhei Imai

Shibaura Institute of Technology
ma20009@shibaura-it.ac.jp

Hiroiyuki Manabe

Shibaura Institute of Technology
manabehiroiyuki@acm.org

ABSTRACT

The spread of personal computer-aided fabrication has made it possible for individuals to create things that meet their personal needs. However, it is still not easy for an individual to prototype electronic circuits. Several methods have been proposed for quickly prototyping electronic circuits. Our solution is combining a 3D printer with transfer foil. This paper expands our original circuit printing method so as to fabricate single-sided multi-layer boards. We show that the proposed technique allows a 3D printer and two types of transfer foil to create single-sided multi-layer circuits.

CCS CONCEPTS

• **Human-centered computing** → **Interface design prototyping**; • **Hardware** → *PCB design and layout*.

KEYWORDS

Multi-layer electric circuit, hot stamping, transfer foil, 3D printer, silver paste

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1 INTRODUCTION

Personal fabrication continues to attract a lot of attention [2]. The barriers to personal fabrication have come down thanks to the falling costs of fabrication machinery such as 3D printers and laser cutters. We are now able to create objects by stacking and cutting. Electronic circuits are needed to add advanced functionality to objects and devices, but are difficult to prototype at home. Many methods have been proposed for rapid prototyping of single-sided, single-layer electronic circuit boards, and some methods have been developed for double-sided electronic circuit boards. However, there are material limitations to all those methods. Although methods for constructing electronic circuits on the surface of various materials have been proposed, they are limited to the construction of single-layer electronic wiring and required manual manipulation.

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We focus on single-sided multi-layer circuits on various kinds of materials. Multi-layer circuits allow more complex circuits to be built in a small area. Single-sided multi-layer circuits are similar to double-sided circuit boards but the electric parts are set on only one side. Thus, they are suitable for attachment to existing surfaces. Furthermore, if the circuit can be directly printed on existing surfaces, neither substrate material nor drilling is necessary.

We have already proposed a technique that allows the user to easily build electric circuits on common surfaces by using a 3D printer [4]. This method can be used to construct electronic circuits on the flat surfaces of various materials, but only simple single-layer circuits can be constructed.

In this paper, we extend our previous technique to create single-sided multi-layer circuits. Since the proposed method can print circuits accurately on just one side of the material, it is possible to add electronic circuits easily to the surfaces of various existing objects and devices.

2 RELATED WORK

A typical example of the rapid prototyping of multi-layered electronic circuits is to fabricate double-sided circuit boards using conductive inks on paper [7]. Unfortunately, this method demands the use of special paper.

Several methods have been proposed for constructing electronic circuits on the surface of materials. ObjectSkin constructs electronic circuits on objects with curved surfaces by using a combination of hydraulic transfer technology and conductive ink [3]. Midas creates circuits from copper foil tape formed by a cutting plotter [5]. Segawa et al. proposed a method of attaching circuit patterns on various materials using gold foil and toner transfer paper [6]. These methods can be applied to curved surfaces, and there is a possibility of applying them to single-sided multi-layer circuit boards. However, manual manipulation is required during transfer and taping, and if the position shifts even once during the fabrication process, the circuit will fail. Unlike these techniques, the proposed technique fixes the material until all the printing process is completed, and the printing is automatically and accurately performed by a 3D printer. It eases concerns about fabrication defects due to misalignment.

A method of constructing an electronic circuit on the skin of the human body using a small plotter has been proposed [1]. A thin insulating layer of liquid bandage is applied to the printed area of the skin in order to prevent the electric current from leaking into the human body. Although this method enables accurate printing of electronic wiring on curved surfaces, it is limited to the printing of single-layer electrical paths.

In our method, thin insulating layers are printed using non-conductive transfer foil to prevent short-circuits when interconnecting different conductive layers.

3 PROPOSED TECHNIQUE

Our previous method [4] replaces the nozzle of the 3D printer with a micro-ball caster to form electronic circuits on a flat surface. The micro-ball caster is heated by the 3D printer and used to apply pressure to a transfer foil set on the substrate material; the circuit path desired is simply traced out. It replicates the hot stamping technique but uses a 3D printer. It was confirmed to suit various materials, such as paper, wood, and acrylic board.

The proposal introduced here is an extension of our previous technique and can easily and cheaply build single-sided multi-layer circuit boards. Unlike conventional multi-layer electric circuits, our technique doesn't require any specific substrate material but can directly print the circuit on existing surfaces without any drilling.

The materials required are those used in the previous study (a FDM 3D printer, micro-ball caster, and conductive transfer foil) plus non-conductive pigment transfer foil. After the electronic circuit paths of the lower layer are printed in the same way as in our previous method, the insulating layer electrically isolates the upper circuit paths by forming the appropriate non-conductive transfer foil tracks. These operations can be repeated to yield a multi-layer circuit board. We have already confirmed that small vertical gaps in the insulating layer form practical vias.

4 IMPLEMENTATION

A regular single extruder FDM 3D printer, CREALITY Ender 5 plus, was used. It costs less than 600 USD and is affordable to individuals. The nozzle of a 3D printer is replaced with the tip of a ballpoint pen (ball diameter 1.6 mm), which is used as a micro-ball caster. The conductive transfer foil is made of silver paste, which achieves a higher conductivity than gold transfer foil. The white pigment transfer foil is made of a mixture of titanium dioxide and adhesive resin, and is commonly used to print white color by hot stamping.

The procedure is as follows. First, with the nozzle of the 3D printer replaced by the micro-ball caster, the base material is placed on the build plate and secured with tape. The conductive transfer foil is then placed on top of the base material and also fixed with tape. After that, the micro-ball caster is heated to 200°C by the 3D printer's nozzle heater. When ready to print, it pressed on the foil and moved to form the circuit at 120mm/m. After printing is completed, the user peel off the conductive transfer foil and scrape off the excess transfer foil on the board with a brush. The insulation layer is printed in the same way. However, the temperature of the micro-ball caster while printing this layer is 180°C, and the travel speed is 300mm/m. These printing processes of electrical paths and insulating layer are alternated starting from the lowest layer.

Fig. 1 shows an example of a three-layer electrical circuit board fabricated by the proposed method on a transparent acrylic board. The wires on both ends are spaced 2.54mm apart. Although three paths pass through the same place in the middle, the paths do not short against each other because the insulating layer (width 3 mm, length 30 mm) is formed 4 times. Note that in producing the structure of Fig. 1, pigment transfer foils were printed four times to form one insulating layer in order to ensure full insulation.

Fig. 2a shows an application example of a single-sided two-layer board. Surface mounted LEDs, resistors, a tact switch, and a PIC micro-controller are attached to the printed wiring on a black

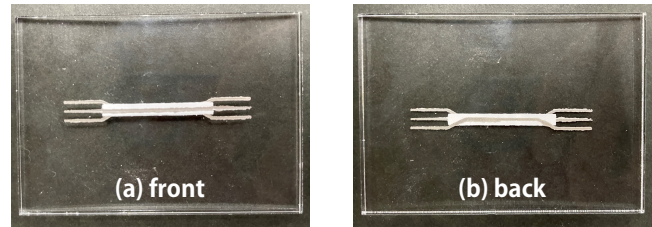


Figure 1: Example of a single-sided three-layer electrical circuit board fabricated by the proposed method.

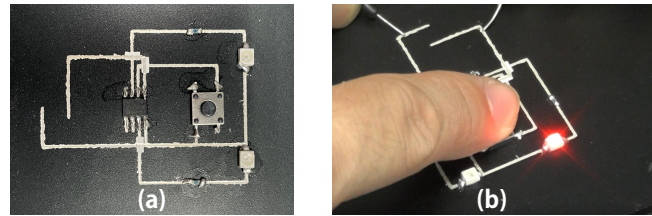


Figure 2: Application example of single-sided two-layer substrate using black acrylic board as base material. (a): entire circuit, (b): How the example application works.

acrylic board using low-melting-point solder. There are three intersections where the insulating layer covers. From Fig. 2b, it can be seen that the circuit properly operates.

5 DISCUSSION

We confirmed the feasibility of the proposed technique. Since our circuits can be directly constructed on the surface of a part or device, electronic wiring can be added to existing objects and devices without compromising the design and in a space-saving manner. However, there are still some challenges with this technique.

Most importantly, we have not conducted fabrication experiments on curved or 3D surfaces. Such tests are essential in order to build electrical circuits directly on various existing objects and devices. In addition, the transfer foil that serves as the insulating layer and the printing method need to be refined. In the application example, the insulating layer was formed by printing on the pigment transfer foil four times, but it is highly desirable to print the transfer foil just once.

6 CONCLUSION

We proposed a technique for making single-sided multi-layer circuit boards using a 3D printer, a micro-ball caster, and 2 types of transfer foils. It is based on our previous technique that uses a hot stamping technique. The proposed technique creates a thin insulating layer made by printing pigment transfer foil between each circuit layer. This method enables the construction of multi-layer electronic circuits on the flat surface of various materials, and thus has the potential to add functions to existing objects and devices at low cost without compromising object design.

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